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Perception of district heating in Europe: A deep dive into influencing factors and the role of regulation

Anna Billerbeck [a,b,*], Barbara Breitschopf [a], Sabine Preuß [a], Jenny Winkler [a], Mario Ragwitz [c,d], Dogan Keles [e]

[a] Fraunhofer Institute for Systems and Innovation Research ISI, Breslauer Straße 48, 76139, Karlsruhe, Germany
[b] University of Freiburg, Faculty of Environment and Natural Resources, Tennenbacher Straße 4, 79106, Freiburg, Germany
[c] Brandenburg University of Technology Cottbus-Senftenberg BTU CS, Platz der Deutschen Einheit 1, 03046, Cottbus, Germany
[d] Fraunhofer Research Institution for Energy Infrastructures and Geothermal Systems IEG, Gulbener Straße 23, 03046, Cottbus, Germany
[e] Technical University of Denmark, Department of Technology, Management and Economics, Produktionstorvet, 424, 2800, Lyngby, Denmark

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ABSTRACT

To increase the deployment of district heating in line with European targets, there is a need to understand consumer perceptions and the role of regulations. Therefore, this paper focuses on consumer perceptions of district heating and analyses the influencing factors. The analysis uses data from an online survey with 4388 participants from nine European countries. In particular, the paper discusses the impact of socio-demographic factors, attitudes and regulations on the perception of district heating. Statistical analyses show that respondents from Denmark and Sweden have the most positive perception, while respondents from Lithuania and the Netherlands have a less positive perception of district heating. In addition, the results indicate that respondents from countries with no mandatory connection, liberalised price regulation and mainly public ownership seem to have a more positive perception of district heating, higher satisfaction when using district heating and a more positive rating of their heating price than those from countries with mandatory connection, regulated prices and a more mixed ownership structure. Overall, the paper provides a first overview of possible factors influencing the perception of district heating and indicates that the mix of appropriate regulations and, in particular, their combined impact, could play an important role in perception.

1. Introduction

The heating and cooling sector accounts for about 50% of European total final energy consumption and is still primarily based on fossil fuels (Eurostat; Heat Roadmap Europe, 2017). Thus, rapid progress in decarbonising the heating and cooling sector is needed to achieve the European Union's (EU) climate neutrality target by 2050 (European Commission, 2020). Given the current energy crisis, decarbonising the heating sector becomes even more important as it reduces the dependence on natural gas. District heating will play a decisive role in decarbonising the heating sector as it allows the use and integration of large shares of renewable and waste heat (Bürger et al., 2019; Connolly et al., 2014). European energy system scenario analyses model strong growth in district heating, with EU market share at least doubling by 2050 (compare overview e.g. in Tsiropoulos et al., 2020).

A characteristic of district heating is the potential for suppliers to exercise excessive market power due to the monopoly-like conditions. Therefore, consumers are shielded by regulations. These regulations are in place to ensure market transparency and consumer rights in order to create trust and acceptance among consumers (Breitschopf et al., 2023; Djørup et al., 2021). At the same time, a variety of approaches towards the regulation of district heating can be observed in the different European countries. These range from stricter, more obligation or rule-based requirements to regulatory frameworks with fewer rules and obligations in place (Bacquet et al., 2022; Billerbeck et al., 2023b). An analysis of the interrelation between regulations and the extent of perceived market power and dependency could reveal how well different regulations contribute to a positive perception of district heating. In other words, if we are to increase the deployment of district heating in alignment with European targets, it will be beneficial to understand consumer perceptions of district heating and the role of regulations.

* Corresponding author. Fraunhofer Institute for Systems and Innovation Research ISI, Breslauer Straße 48, 76139, Karlsruhe, Germany.
E-mail address: anna.billerbeck@isi.fraunhofer.de (A. Billerbeck).

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This forms the starting point of this paper. Our contribution focuses on consumers' perception and satisfaction with district heating and identifies influencing factors, including different regulations. While there are a few studies on acceptance issues in the area of district heating, there is, to our knowledge, no scientific paper on the perception of district heating encompassing several European countries with a comparative discussion of them. There are, to our knowledge, a few project reports providing insights into the topic (Bacquet et al., 2022; Breitschopf et al., 2023) and a few scientific papers analysing perceptions or the willingness to adopt district heating at the country level (e.g. Denmark (Skytte and Olsen, 2016), Germany (Krikser et al., 2020; Zaunbrecher et al., 2016), Sweden (Mahapatra and Gustavsson, 2009)). However, there is no analysis across countries, which looks into a variety of factors and the role of regulations. Our aim is to fill this gap by addressing the following questions: Which factors influence the perception of district heating in the different countries? Do regulations influence the perception of district heating?

The analysis covers nine countries: Denmark, France, Germany, Italy, Lithuania, Netherlands, Poland, Slovakia and Sweden. This selection takes into account an equal distribution of countries based on geographical location and climate, economic strength, share of district heating as well as share of renewable and waste heat in district heating. To assess the country-specific regulations, we conduct a literature review and collect current district heating regulations. Based on the existing regulations, we assign the countries to different groups. To assess the perception of district heating, we use data from an online household survey in the nine countries, conducted in the project 'Overview of Heating and Cooling: Perceptions, Markets and Regulatory Frameworks for Decarbonisation', in short 'Perception' (Breitschopf et al., 2023). The sample size of this survey comprises 4388 participants. Finally, we discuss the impact on the perception of district heating of socio-demographic factors, attitudes and regulations. Methodologically, we use group comparisons with test statistics and for verification we conduct a regression analysis using ordinal logistic regression models.

The paper is structured as follows: Section 2 provides a theoretical background on relevant district heating regulations. In section 3, the data and methodology used are described. The results are presented in section 4 and discussed in section 5. We conclude with policy implications in section 6.

2. District heating regulations

Unlike electricity and natural gas grids, district heating is often a vertically integrated service in which generation, grid and distribution are operated in an integrated manner by one company (Bacquet et al., 2022). Thus, institutional settings and regulations are established to ensure market transparency and protect consumers from abuse. Regulation varies, ranging from stricter, more rule-based requirements like in Denmark to frameworks with simpler or fewer rules, such as the Finnish competition-based regulation (Bacquet et al., 2022; Billerbeck et al., 2023b; Sandberg et al., 2018). According to Djørup (2021), the regulation of district heating should consider a company perspective and a societal perspective. From a company perspective, the regulations should address the high upfront capital costs, which the long-term investment perspective of district heating necessitates, the associated risks and the access to capital. From a societal perspective, the regulation should deliver consumer acceptance and protection as well as the ability to support long-term planning to reach climate targets.

In terms of consumer perceptions, a variety of regulations may be relevant. The regulations, which directly affect the consumer, are those for grid connection and usage (i.e. mandatory consumer connection) and the regulation of prices, as these regulations determine whether consumers are forced to use district heating and what they have to pay. In addition, the ownership structure of consumers' local district heating network is important for consumers as it determines their ability to participate and influence decisions. We, therefore, address these three aspects in the following. Furthermore, in the results section, we use this (theoretical) background to describe the regulations and then to group countries with similar regulations together in order to analyse the statistical differences between the groups.

2.1. Mandatory consumer connection

District heating systems are often faced with the challenge of not enough citizens being motivated to connect to their network, leading to low profitability of the systems (Bacquet et al., 2022). To address this, some countries or regions have introduced mandatory grid connection and usage for consumers, which means that consumers are obliged to connect to their local district heating system under certain conditions (Bacquet et al., 2022). In most cases, such an obligation applies only to new buildings in certain regions (Bacquet et al., 2022). Furthermore, mandatory connection and usage for the consumer must be well justified, i.e. it can only be implemented when a district heating network fulfils certain criteria like high shares of renewables or high cost efficiency (Bacquet et al., 2022). Mandatory grid connection may have an impact on the acceptance of district heating. Consumers may feel locked in, as they are not free to choose their heating system or the heat supplier (compare e.g. Hodges et al., 2018).

On the one hand, mandatory connection gives more security, especially investment security, to district heating companies and can therefore lead to lower prices. On the other hand, mandatory connection leads to even greater consumer dependency, which increases the importance of effective regulation. In particular, price regulation is becoming more important as in such cases consumers are obliged to buy heat regardless of the price. Thus, there is a trade-off between the company perspective and the societal or consumer perspective, as described by Djørup (2021). With this in mind, we group the nine countries in the result section according to whether they have implemented mandatory consumer connection or whether they do not rely on such an obligation.

2.2. Price regulation

As discussed in several analyses (e.g. Bacquet et al., 2022; Billerbeck et al., 2023a; Billerbeck et al., 2023b), the formation of district heating prices differs between countries and depends primarily on the principles by which the district heating sector is regulated. District heating prices may be regulated or liberalised, and prices may be subject to price controls (Bacquet et al., 2022). Regulated pricing refers to certain price calculation rules that are often combined with price controls, while liberalised pricing means that prices are formed more or less freely based on the costs of the supplier and on the market conditions (Bacquet et al., 2022). Price control can be ex-ante or ex-post as described by Wissner (2014). In the case of ex-ante regulation, prices have to be approved by an authority before they are applied. In the case of ex-post control, the authority checks ex-post whether the prices were in accordance with the established price rules. Billerbeck et al. (2023b) categorise price regulation into three different approaches: (1) no specific regulation, i.e. only cartel law; (2) price is defined by a district heating operator with a price calculation rule and (3) price is defined or set by the regulator or defined by the operator with ex-ante price control.

Furthermore, Odgaard and Djørup (2020) define seven regulatory principles for district heating pricing: (1) The 'true-cost principle' which implies that the consumer price equals all necessary costs of production and distribution. (2) The 'regulated return to investor principle' implies that the price is based on the true costs, as described above, but higher loan costs are allowed, thus, external investors can provide credit based on market conditionality. (3) The 'prices set by the market' option means full liberalisation of district heating consumer prices, which are based on supply and demand. (4) The 'substitution price' links the consumer price to the same level as heat produced by individual heating systems (e.g. natural gas boilers). (5) The 'price cap' sets a maximum price, mostly according to the heating price of alternative supply. (6) The principal 'private operation under public ownership' means that local governments may own district heating without operating it. Prices or price rules might be defined in the contract under which the network is transferred. (7) Finally, the 'energy service company model' means that an energy service company can provide the required investment capital, technology and information to establish a network. Typically, heat is sold for a fixed price several years in advance.

In conclusion, district heating price regulations vary in their overall approach and in their intensity. The price regulations have an impact on end consumer prices, and in turn, prices influence the perception of district heating, as shown by Zaunbrecher et al. (2016). We use the regulatory principles for district heating pricing to describe price regulation in the nine countries (compare Table A5 in the appendix). Furthermore, we group the countries in the result section according to their overall approach to district heating pricing, i.e. regulated pricing vs. more liberalised pricing.

2.3. Ownership structure

In most countries, district heating networks are operated by vertically integrated companies, owned by varying forms of private or public companies or by consumer cooperatives (Bacquet et al., 2022; Billerbeck et al., 2023b). Based on these real-world ownership models, Djørup et al. (2021) define three different forms of ownership: (1) private ownership, (2) public ownership and (3) consumer ownership. They discuss the dimensions of consumers' power. Private ownership refers to a district heating network owned and operated by a private company and consumer power is low. In turn, the company is regulated by the state to prevent the abuse of market power. Public ownership refers to a district heating network that is operated by a company that is owned by a municipality. Thereby the municipality has a certain power over the company because of its ownership. In turn, consumers have some democratic influence on the municipality. Finally, consumer ownership means that the consumers own the company that operates their local district heating system and thus consumers have direct power over the company. Fig. 1 illustrates the three ownership models and the relative dimension of consumer power.

Biggar and Söderberg (2020) and Egüez (2021) analyse correlations between ownership and prices, focusing on the Swedish district heating market. They show that public district heating networks have lower prices compared to privately owned networks. Similarly, Djørup et al. (2021) conclude from their analysis that consumer ownership of district heating networks plays an important role in the success of price regulation, and, thus, in consumer trust and acceptance of district heating in Denmark. For the analysis, we describe the ownership of district heating in the nine countries and group countries in the result section according to their ownership structure, i.e. mainly public vs. higher share of private players.

In summary, mandatory consumer connection, price regulation and the ownership structure of district heating networks may affect consumer perceptions. Therefore, we analyse the impact of these regulations on consumer perception, which has not been done before. With this analysis, we contribute to a better understanding of consumer perception of district heating and the role of regulation, which can help to promote the further deployment of district heating.

3. Data and methods

3.1. Data collection process and data

To analyse the perception of district heating in the context of institutional settings and regulation, we collect data on real-world regulations and institutional settings for district heating in the nine selected countries. We use the regulatory principles described in section 2 to describe regulations and to sort countries into groups according to the similarity of their regulations and institutional settings.

To assess perceptions of district heating, we use data from an online household survey conducted in the project 'Perception' (see acknowledgements and Breitschopf et al., 2023). The survey contained around 30 questions and was structured around the following topics: (1) Socio-demographic characteristics, (2) current heating system, (3) factors influencing decisions in heating and cooling, (4) perception of district heating, (5) attitudes, values and trust in promoters. The exact questions analysed in this paper are presented in Table A1 in the appendix. The recruitment of participants was supported by a market research institute and participation in the survey was possible from June to August 2022. Consequently, the survey results might be affected by the energy crisis following the COVID-19 pandemic in 2021 and Russia's invasion of Ukraine in 2022.

The survey covered nine countries: Denmark, France, Germany, Italy, Lithuania, Netherlands, Poland, Slovakia and Sweden. The selection of countries took into account an equal distribution of countries based on geographical location and climate, economic strength, share of district heating and share of renewables and waste heat in district heating. To receive representative samples, gender, age, education, household size and share of district heating were applied as quotas. To ensure high data quality, we implemented two control items and

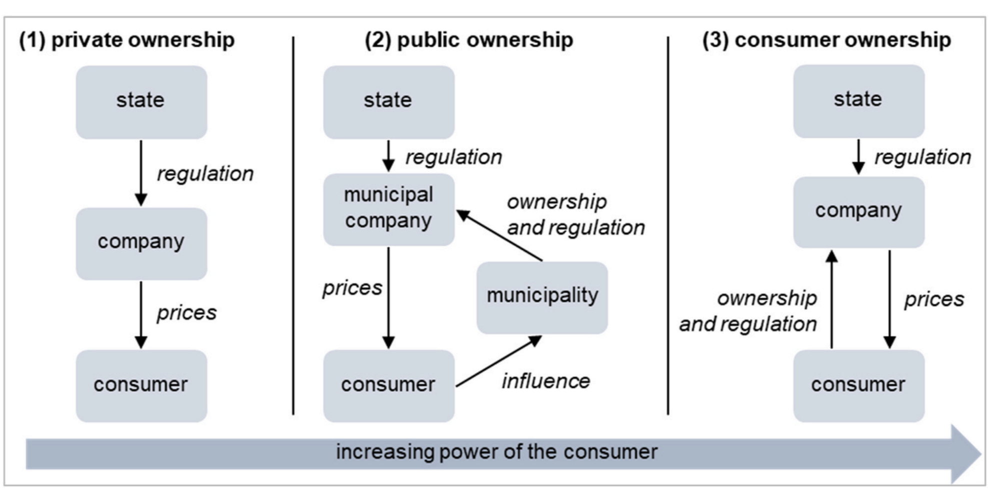

Fig. 1. Ownership models of district heating systems (building on Djørup et al., 2021; Djørup, 2021) *(Note: Consumers may not be the direct customers of the district heating company. Particularly in multi-family houses with tenants, the end user is rarely the direct customer. In these cases, the building owner, often represented by a property management company, is the customer of the district heating company.)*

Table 1
Overview of samples per country.

Country	n	DH user	DH non-user	Share female	Median age	Share tertiary education	Average household size
Denmark	500	284	216	50%	53	68%	1.93
France	491	48	443	50%	54	59%	2.23
Germany	493	73	420	50%	56	69%	1.99
Italy	513	38	475	51%	48	35%	2.29
Lithuania	450	183	267	66%	45	86%	2.55
Netherlands	476	36	440	52%	52	67%	2.12
Poland	505	195	310	57%	41	53%	2.73
Slovakia	466	102	364	54%	43	34%	2.93
Sweden	494	222	272	52%	54	56%	2.01

(n = number of respondents, DH = district heating)

Table 2
Overview of variables.

Category	Variable	Type	Values and ranges
Perceptions	perception	ordinal	1 (very negative) to 5 (very positive)
	satisfaction	ordinal	1 (very dissatisfied) to 5 (very satisfied)
	price rating	ordinal	1 (very high) to 5 (very low)
Socio-demographic factors	gender	categorical	female, male, other
	age	metric	open ranging from 18 to 89
	education	categorical	1 (primary), 2 (secondary), 3 (tertiary), 4 (academic degree)
Heating system	user	categorical	user vs. non-user
Attitudes and values	eco awareness	metric	average of three categorical variables with 1 (I totally agree) to 5 (I totally disagree)
	technology affinity	metric	
	trust in policymakers	metric	
Regulation	mandatory connection	categorical	no mandatory connection vs. mandatory connection
	price regulation	categorical	regulated vs. liberalised price regulation
	ownership structure	categorical	mainly public vs. mixed ownership

excluded respondents from the analyses based on the following criteria: (1) Respondent is not responsible for energy decisions within the household; (2) Data quality is low, i.e. both control items were answered incorrectly; (3) No answer to the country question; (4) Age under 18; (5) Screen out due to the set quotas; (6) Respondents who completed the survey in less than 90 s. Overall, 8756 responses were received, but around 50% (4368) were excluded due to the criteria. This led to a final sample size of 4388 participants. Table 1 briefly describes the sample per country. An overview of the extent to which the sample met the quotas in each country is presented in Table A2 in the appendix.

The variables used in the analysis are presented in Table 2. In the survey, perception was captured by the question 'How do you rate district heating as a heating system overall?' and expressed on a 5-point Likert-type scale from 'very negative' to 'very positive'. Similarly, satisfaction of the actual users of district heating was rated and expressed on a 5-point Likert-type scale. Price rating was captured with 'How would you describe your price for heating with district heating (in comparison with your income)?' and again expressed on a 5-point Likert-type scale from 'very high' to 'very low'.

To identify the possible relationships between perception and influencing factors, we consider a number of factors. Firstly, socio-demographic factors such as gender, age and education were collected for each respondent via the online survey, mainly as control variables. Secondly, the use of district heating as a heating system in the household was captured for each respondent via the survey. Thirdly, the attitudes and values of each respondent in terms of environmental awareness (eco awareness), technology affinity and trust in policymakers were recorded, as these are potential factors influencing the perception (compare e. g. Zaunbrecher et al., 2016). To capture the attitudes, participants were asked to indicate their agreement or disagreement with statements such as 'Acting in an environmentally friendly way is an important part of who I am' (see Table A1 in the appendix for all statements). The attitude variables are an average of agreement or disagreement with three statements each. Finally, and in line with the overall research question, regulatory issues related to district heating were collected using a literature review. These variables are at the country level and are explained in more detail in section 4.3.

3.2. Methodology

As stated in section 3.1, survey participants were asked to signal their agreement or disagreement to statements or factors by using 5-point Likert-type scales. Likert scale data can be analysed as ordinal data. Methodologically, we use (i) descriptive statistics and as verification, we conduct a (ii) regression analysis using ordinal logistic regression models. Details are described in the following. To compute the tests and conduct the regression analysis, we used the programming language R ('polr' from 'MASS').

(i) Descriptive statistics

In the descriptive result section, we use the distribution of responses, mean m and median $\widetilde{m}$ to show tendencies (compare sections 4.1, 4.2 and 4.3). In particular, we discuss the differences between countries and the impact of socio-demographic factors, attitudes and regulations using group comparisons. We apply a Mann-Whitney-U-test, which can be used to test whether there is a statistically significant difference between two independent groups when the data are not necessarily normally distributed.

The Mann-Whitney-U-test is based on the idea of ranking the data (see e.g. Bauer, 1972; Hollander et al., 2014; Universität Zürich, 2023). Rank sums are formed for both groups. To calculate the tests, i.e. the Mann-Whitney-U-value, the rank sum R_1 of the first group with n_1 obseravtions is used as follows:

$$U = R_1 - \frac{n_1(n_1 + 1)}{2} \quad (1)$$

To assess the relevance of a result, we calculate effect sizes and p-values (significance threshold of 0.05). For the effect size, we use the Pearson correlation coefficient r (i.e. correlation of the perception of the two groups through pairwise comparison) (R Documentation; Universität Zürich, 2023). To calculate the correlation coefficient the z-score and the sample size n are used (compare formula 2). The z-score is the z-standardised U value, calculated with the mean of the distribution μ_U and the standard deviation σ_U (see formula 3).

$$r = \left| \frac{z}{\sqrt{n}} \right| \tag{2}$$

$$z = \frac{U - \mu_U}{\sigma_U} = \frac{U - \frac{n_1 \cdot n_2}{2}}{\sqrt{\frac{n_1 \cdot n_2 \cdot (n_1 + n_2 + 1)}{12}}} \tag{3}$$

(ii) Regression models

In addition to the above testing scheme, we fit a total of three regression models to assess variation in perception with respect to the factors described earlier (compare Table 2). As a modelling framework, we use ordinal logistic regression (see e.g. McCullagh, 1980; Qiao et al., 2021). Since we analyse ordinally scaled data as the dependent variable, ordinal logistic regression is appropriate for data analysis. Given an ordinal outcome Y with J categories, the model targets the logit of $P(Y \le j)$, i.e. the cumulative probability of Y being less than or equal to a specific category j. The full specification of model 1 is then:

$$\log\left(\frac{P(perception_i \le j)}{1 - P(perception_i \le j)}\right) = \alpha_j - (\beta_1 gender_i + \beta_2 age_i + \beta_3 education_i + \beta_4 user_i + \beta_5 eco\ awareness_i + \beta_6 technology\ affinity_i + \beta_7 trust\ in\ policymakers_i) \tag{4}$$

Note that this model parametrization uses a negative sign linking the level-specific intercept α_j and the predictor terms. This specification enforces that a positive coefficient indicates that as the value of the explanatory variable (here: perception) increases, the likelihood of a higher ranking increases.

Model 2 contains the same predictors as model 1 but additionally covers fixed country effects. Similarly, model 3 contains the same predictors as model 1 but also includes the country-level regulation variables *mandatory connection*, *price regulation*, *ownership structure* (compare Table 2). In the models, all metric predictors are centred and scaled by subtracting the mean and dividing by the standard deviation.

4. Results on the perception of district heating

4.1. Overall perception in the countries

Descriptive statistics show that European households seem to have a positive perception of district heating, as the majority of the respondents (52%) rated district heating as positive or even very positive, while only a smaller number of respondents (10%) rated district heating as negative or very negative. Fig. 2 shows the perception in the nine different countries. The perception of district heating is particularly positive in Denmark, followed by Sweden, Poland, Italy and Germany. In the Netherlands and in Lithuania, the perception is less positive, with around 20% rating district heating as negative or very negative. However, the mean is still relatively high, indicating a neutral to positive image on average. In comparison, France and Slovakia are in the middle range.

Mann-Whitney-U-tests were calculated to determine if there were significant differences between the countries (compare methodology in section 3). The results of the test statistics are shown in Table 3. Thereby, Denmark was used as a reference group as it shows the most positive perception. Thus the test statistics show the difference between Denmark and the respective country. The results show that there are

Table 3
Test statistics of perception in the countries with Denmark as a reference group (own calculations, data from Breitschopf et al., 2023).

Variable	Mann-Whitney-U-value	Effect size r	p-value
France	167093	0.32 (medium)	0.000
Germany	152202	0.22 (small to medium)	0.000
Italy	156323	0.20 (small to medium)	0.000
Lithuania	162122	0.40 (medium)	0.000
Netherlands	177595	0.45 (medium)	0.000
Poland	152079	0.19 (small)	0.000
Slovakia	159401	0.36 (medium)	0.000
Sweden	146611	0.17 (small)	0.000

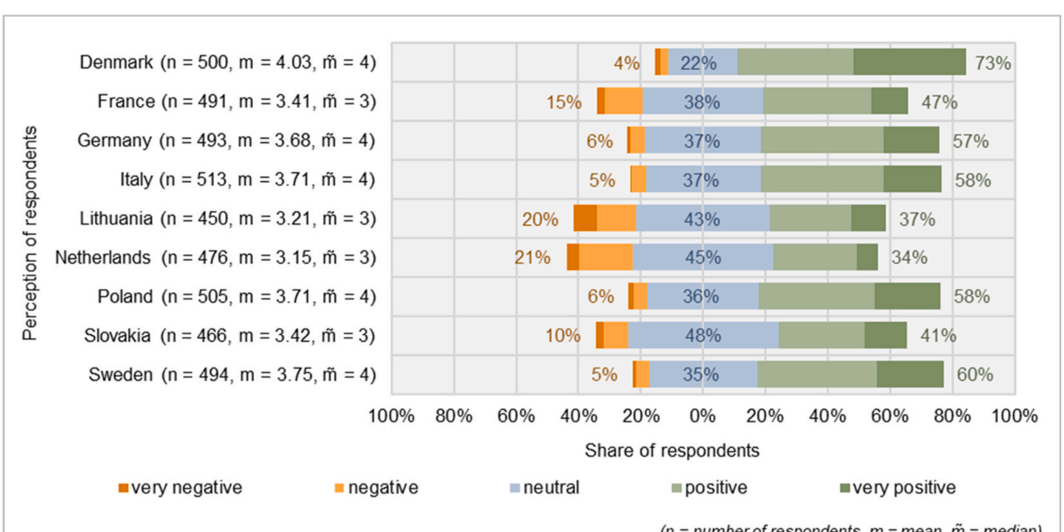

Fig. 2. Perception of district heating in the different countries (own calculations, data from Breitschopf et al., 2023).

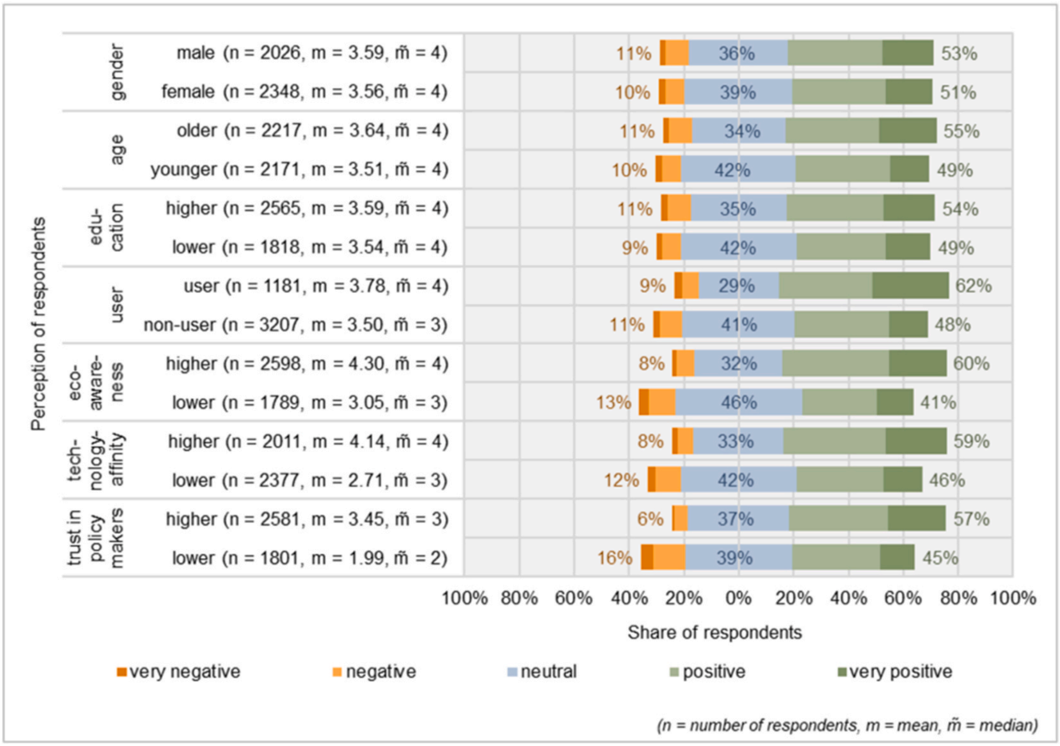

Fig. 3. Perception of district heating in different groups (own calculations, data from Breitschopf et al., 2023).

small to medium, statistically significant differences between the countries (compare U-value, effect size r and p-value in Table 3). The biggest difference can be observed between Denmark and the Netherlands, followed by Lithuania, Slovakia and France.

Regarding the users' satisfaction with district heating, the descriptive analyses show that the overall picture is similar to the overall perception in the countries (compare Table A3 in the appendix). The highest satisfaction regarding the use of district heating can be observed in Sweden, followed by Germany, Poland and Denmark. In the Netherlands and Lithuania, the respondents are again less satisfied. However, on average, high satisfaction was also reached in those countries. France, Italy and Slovakia are again in the middle range.

4.2. Impact of socio-demographic factors and attitudes

Fig. 3 shows the perception of district heating from different groups, divided by socio-demographic factors and attitudes as well as users vs. non-users. Again, we compared the descriptive statistics of the groups by using Mann-Whitney-U-tests. Table 4 shows the test statistics for these group comparisons. Regarding gender, differences are not statistically significant. However, statistically significant differences can be observed regarding age (mean split) and education ('low' relates to primary and secondary education, while 'high' relates to tertiary education and academic degree). The results show that older respondents perceive district heating significantly better than younger respondents. However, the effect is very small. Furthermore, people with higher education have significantly better perceptions of district heating than people with lower levels of education. However, the effect is so small that it should be treated as no effect. Moreover, users have a significantly better perception of district heating compared to non-users. The effect is again rather small.

In addition to these socio-demographic standard variables, differences expressed in terms of environmental awareness (i.e. eco awareness), affinity to (new) technologies and trust in policymakers are assessed. Again, a mean split was performed to create the groups. The comparison shows that respondents with a higher eco awareness (i.e. above average) have a significantly better perception of district heating compared to respondents with lower eco awareness. The same applies to respondents with a higher affinity to (new) technology compared to their counterparts. The largest difference can be observed between respondents with a higher trust in policymakers (good perception) and respondents with less trust in policymakers (worse perception of district heating). The differences are all significant but the effect sizes are small. However, it is noteworthy that the effect sizes for these groups are larger than the effects of the socio-demographic factors. Hence, attitudes seem to have a stronger influence on the perception of district heating than socio-demographic factors.

We performed the same group comparison for each country (compare Table 5). The results show that the overall picture is similar: In (almost) all countries attitudes have a stronger influence on the perception of district heating than socio-demographic factors. Thereby, the direction of the effects in each country is the same as for the whole sample (compare means and medians in Table A4 in the appendix). In particular, eco awareness has a small to medium effect in about half of the countries. However, there are also clear differences between the countries. In Denmark, eco awareness seem to have the highest impact

Table 4
Test statistics of perception in the context of socio-demographic factors and attitudes (own calculations, data from Breitschopf et al., 2023).

Variable	Mann-Whitney-U-value	Effect size r	p-value
gender	2339289	–	0.321
age	2218552	0.07 (very small)	0.000
education	2236235	0.04 (very small)	0.014
user	1559329	0.14 (small)	0.000
eco awareness	1852583	0.18 (small)	0.000
technology affinity	2009634	0.15 (small)	0.000
trust in policymakers	1904238	0.17 (small)	0.000

Table 5
Test statistics of perception in the context of socio-demographic factors and attitudes per country (own calculations, data from Breitschopf et al., 2023).

Country	Variable	Mann-Whitney-U-value	Effect size r	p-value
Denmark	gender	30249	–	0.515
	age	**21417**	**0.29 (small to medium)**	**0.000**
	education	22037	0.17 (small)	0.000
	user	24564	0.18 (small)	0.000
	eco awareness	**20709**	**0.26 (small to medium)**	**0.000**
	technology affinity	29705	–	0.342
	trust in policymakers	24701	0.17 (small)	0.000
France	gender	29837	–	0.845
	age	29870	–	0.960
	education	30539	–	0.373
	user	10425	–	0.815
	eco awareness	**19118**	**0.26 (small to medium)**	**0.000**
	technology affinity	**20189**	**0.29 (small to medium)**	**0.000**
	trust in policymakers	22994	0.19 (small)	0.000
Germany	gender	30717	–	0.821
	age	26411	–	0.300
	education	24426	–	0.438
	user	11682	0.16 (small)	0.000
	eco awareness	**21091**	**0.22 (small to medium)**	**0.000**
	technology affinity	27507	–	0.059
	trust in policymakers	25025	0.13 (small)	0.005
Italy	gender	33494	–	0.679
	age	28218	0.13 (small)	0.003
	education	28057	–	0.171
	user	9363	–	0.683
	eco awareness	**24024**	**0.25 (small to medium)**	**0.000**
	technology affinity	26402	0.18 (small)	0.000
	trust in policymakers	27263	0.10 (small)	0.029
Lithuania	gender	20987	–	0.146
	age	24393	–	0.482
	education	11276	–	0.657
	user	21749	0.10 (small)	0.037
	eco awareness	21989	–	0.077
	technology affinity	25775	–	0.710
	trust in policymakers	**20112**	**0.17 (small)**	**0.000**
Netherlands	gender	28559	–	0.833
	age	33592	0.17 (small)	0.000
	education	23365	–	0.229
	user	8076	–	0.835
	eco awareness	20826	0.18 (small)	0.000
	technology affinity	25235	–	0.183
	trust in policymakers	**20414**	**0.24 (small to medium)**	**0.000**
Poland	gender	32174	–	0.500
	age	27048	0.13 (small)	0.003
	education	31396	–	0.832
	user	26733	0.10 (small)	0.021
	eco awareness	**22797**	**0.20 (small to medium)**	**0.000**
	technology affinity	26089	0.15 (small)	0.000
	trust in policymakers	25956	0.12 (small)	0.007
Slovakia	gender	27722	–	0.583
	age	26553	–	0.680
	education	22824	–	0.262
	user	17697	–	0.438
	eco awareness	22521	0.15 (small)	0.001
	technology affinity	22592	0.14 (small)	0.002
	trust in policymakers	**21190**	**0.17 (small)**	**0.000**
Sweden	gender	28532	–	0.202
	age	25497	0.15 (small)	0.001
	education	24522	0.16 (small)	0.000
	user	24342	0.18 (small)	0.000
	eco awareness	24577	0.16 (small)	0.000
	technology affinity	**23384**	**0.21 (small to medium)**	**0.000**
	trust in policymakers	23816	0.13 (small)	0.003

(Note: Highest effect sizes are presented in bold. Furthermore, it should be noted that in France, Italy and the Netherlands the sample obtained includes only a small number of district heating users, which makes it difficult to obtain significant results on country-specific issues for users in these three countries.)

(comparing the effect sizes), while in Sweden technology affinity has the strongest effect compared to the other factors. Furthermore, in Lithuania, the Netherlands and Slovakia, trust in policymakers shows the highest effects. Socio-demographic variables seem to play a role in some countries but not in others: In Denmark, Poland, the Netherlands and Sweden an effect for age occurs, whereas in France, Germany, Lithuania and Slovakia, the perception of district heating does not differ between age groups. Lastly, we do not find gender-related differences playing a role in the perception in any country.

4.3. *Role of regulation and institutional settings*

To analyse the perception of district heating in the context of regulations and settings, we collect information on real-world regulations and settings for district heating in the nine countries. In doing so, the theoretical characteristics of regulations, described in section 2, are applied. Details on the regulation and the ownership structure in each country are outlined in Table A5 in the appendix. Table 6 summarises

Table 6
Overview of regulations and ownership structure in the different countries (Bacquet et al., 2022; Billerbeck et al., 2023b; Djørup et al., 2021).

Country	Consumer grid connection	Price regulation	Ownership structure
Denmark	mandatory connection (until 2019)	regulated	mainly public (with consumer cooperatives)
France	mandatory connection	liberalised	mainly public (and PPP)
Germany	mandatory connection	liberalised	mainly public
Italy	no mandatory connection	liberalised	mixed
Lithuania	mandatory connection	regulated	mainly public
Netherlands	mandatory connection (very rarely)	regulated	mixed (with consumer cooperatives)
Poland	no mandatory connection	regulated	mixed
Slovakia	mandatory connection	regulated	mixed
Sweden	no mandatory connection	liberalised	mainly public

(Note: Mandatory connection addresses in most cases only specific areas or buildings and not all consumers; PPP = public-private partnership) the collected information in the nine countries. This summary shows that the majority of countries have some form of mandatory connection for consumers (i.e. for certain areas or buildings). Price regulation varies, with five countries having a more regulated approach to district heating prices and four countries having a more liberalised form of price regulation. Similarly, there are five countries where district heating is mainly publicly owned (i.e. a share of more than 50%, compare Table A5 in the appendix) and four countries with a more mixed ownership structure, i.e. also a significant share of private owners. However, private ownership is less common overall, thus we do not define the category of 'predominantly private ownership'.

To analyse the perception of district heating in the context of regulations, we sort countries into groups according to the similarity of their approaches (as presented in Table 6). We then examine potential group differences by comparing country groups based on (i) mandatory consumer connection (mandatory vs. no mandatory connection), (ii) price regulation (regulated vs. liberalised pricing) and (iii) ownership structure (public vs. mixed ownership). This time we do not only focus on the perception of district heating as a dependent variable but also include a more detailed analysis of the perspective of district heating users. We show results for the perception of all respondents, the perception of (only) users of district heating, as well as the satisfaction of users and the price rating of users.

(i) Mandatory consumer grid connection

Regarding consumer grid connection, the countries are divided into (1) countries where an obligation for consumers to connect to their local district heating network is in place, i.e. mandatory connection and usage under certain conditions and/or in specific areas (Denmark, France, Germany, Lithuania, Netherlands, Slovakia) and (2) countries without mandatory connection (Italy, Poland, Sweden). Fig. 4 shows the perception, satisfaction and price rating per group. In this comparison, respondents from countries *without* mandatory connection show a more positive perception and a higher level of satisfaction with district heating. Furthermore, users of district heating from this country group rated their price for district heating as more positive (i.e. lower price) than respondents from countries with mandatory connections. Conducting Mann-Whitney-U-tests shows that the group differences are significant, leading to small effects regarding the perception of all respondents as well as the satisfaction and price rating of users (compare Table 7). The analysis of the difference in users' perceptions shows no significant effect.

Table 7
Test statistics of perceptions in the context of mandatory connection (own calculations, data from Breitschopf et al., 2023).

Variable	Mann-Whitney-U-value	Effect size r	p-value
Perception (all)	2453816	0.11 (small)	0.000
Perception (user)	172589	–	0.173
Satisfaction (user)	186423	0.11 (small)	0.000
Price rating (user)	178957	0.08 (very small)	0.009

(ii) Price regulation

For price regulation, the countries are divided into (1) countries with regulated prices (Denmark, Lithuania, Netherlands, Poland, Slovakia) and (2) countries with more liberalised prices (France, Germany, Italy, Sweden). Fig. 5 shows the descriptive statistics indicating that respondents from the countries with more *liberalised* price regulation show a more positive perception and a higher level of satisfaction with district heating. Furthermore, users of district heating from this country group rated their price for district heating more positively than respondents from countries with more regulated prices, i.e. respondents in group (2) seemed to perceive their price for district heating as lower compared to the other group. The Mann-Whitney-U-tests show that the group differences are significant with very small effects regarding perception and satisfaction, and with small to medium effects regarding the price rating (compare Table 8). Again, looking only at users' perception, the test statistic shows no significant effect. The small to medium effect on price

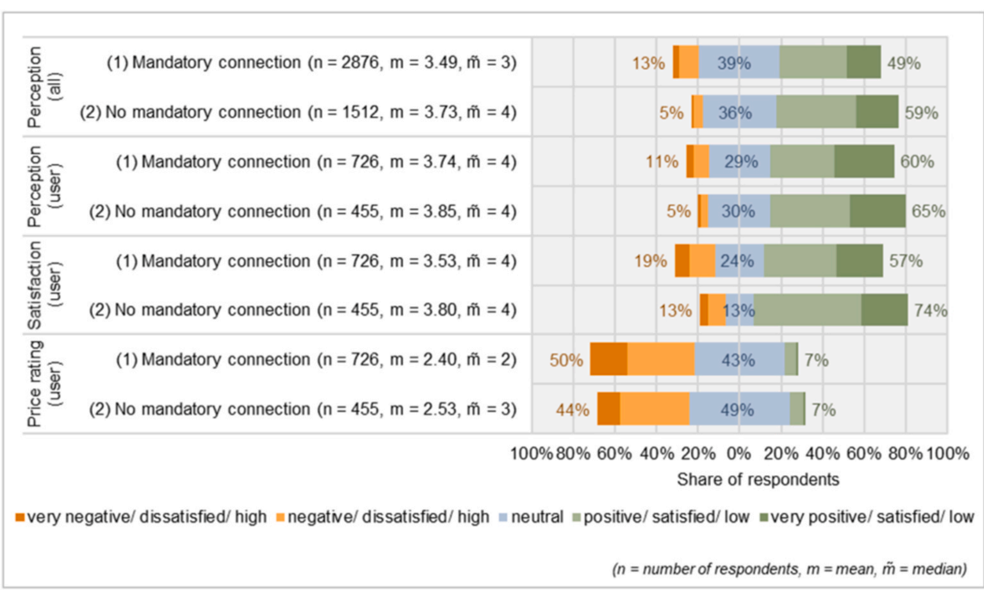

Fig. 4. Perception in the context of mandatory connection (own calculations, data from Breitschopf et al., 2023).

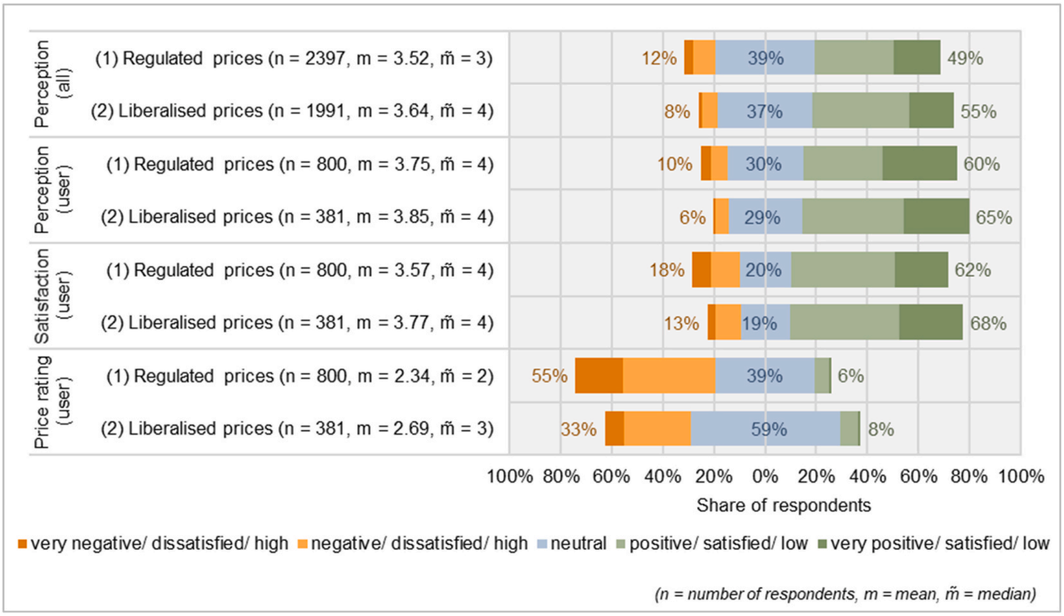

Fig. 5. Perception in the context of price regulation (own calculations, data from Breitschopf et al., 2023).

Table 8
Test statistics of perceptions in the context of price regulation (own calculations, data from Breitschopf et al., 2023).

Variable	Mann-Whitney-U-value	Effect size r	p-value
Perception (all)	2231994	0.06 (very small)	0.000
Perception (user)	147700	–	0.370
Satisfaction (user)	138595	0.08 (very small)	0.008
Price rating (user)	117116	0.20 (small to medium)	0.000

rating can be explained by the fact that price regulation impacts prices and thus the rating of the prices.

(iii) Ownership structure

Finally, with regards to ownership structure, the countries are divided into (1) countries with mainly public ownership (including consumer ownership) and/or public-private partnership (Denmark, France, Germany, Lithuania, Sweden) and (2) countries with a mix between public and private ownership of district heating, i.e. a substantial share of private players (Italy, Netherlands, Poland, Slovakia). Fig. 6 shows that respondents from the countries with mainly *public* ownership of district heating show a more positive perception of district heating. This applies for users and all respondents. Furthermore, users of district heating from this country group seemed to perceive their price for district heating as lower compared to the other group. The Mann-Whitney-U-tests show that the group differences are significant with (very) small effects regarding the perception of district heating and small to medium effects regarding the price rating (compare Table 9). For users, the test statistic of satisfaction shows no significant effect.

In summary, the results show that there are significant differences between the groups on the basis of regulations, although the effects observed are only small to medium. Concluding on these results, we find

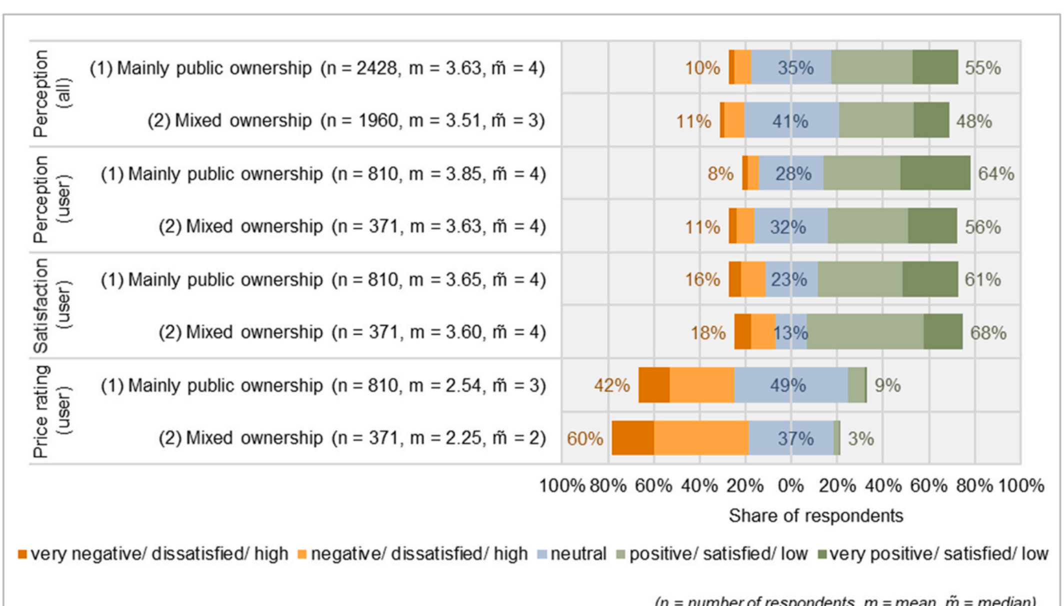

Fig. 6. Perception in the context of ownership structure (own calculations, data from Breitschopf et al., 2023).

Table 9
Test statistics of perceptions in the context of ownership structure (own calculations, data from Breitschopf et al., 2023).

Variable	Mann-Whitney-U-value	Effect size r	p-value
Perception (all)	2561649	0.07 (very small)	0.000
Perception (user)	168645	0.10 (small)	0.000
Satisfaction (user)	152574	–	0.655
Price rating (user)	178775	0.16 (small to medium)	0.000

that respondents from countries with no mandatory connection, liberalised price regulation and mainly public ownership have a more positive perception of district heating, higher satisfaction with the use of district heating and a more positive rating of their district heating price.

4.4. Econometric results

In this section, the results from the three ordinal logistic regression models are presented (compare methodology in section 3.2). The overall aim of the regression models is to verify and validate the tendencies identified with the test statistics (compare sections 4.1, 4.2 and 4.3). Hence, we use the models to check the robustness and to jointly assess the impact of socio-demographic factors, attitudes and regulations on the perception of district heating in the different countries.

Summary statistics for the variables used in the models are presented in Table 10. Perception of district heating presents the dependent variable, while the other variables are used as predictors in the model. The correlation between predictors is low (compare Table A6 in the appendix). The selection of the predictors is based on existing literature, the preceding analyses on influencing factors and attitudes and the role of regulations.

Table 11 presents the coefficients, errors, odd ratios (OR) and p-values of the ordinal logistic regression models built to assess the impact of the variables on the perception of district heating. Model 1 focuses on the impact of socio-demographic factors and attitudes on the perception of district heating, while model 2 also includes fixed country effects and model 3 additionally includes the defined regulation variables (compare

Table 10
Summary statistics for variables (own calculations, data from Breitschopf et al., 2023).

Variable	type	n	mean	median	SD	min or level	max or level
perception	ordinal	4388	3.57	4	0.95	1 (very negative)	5 (very positive)
gender	categorical	4388	–	–	–	female	male
age	metric	4388	48.40	49	16.12	18	89
education	categorical	4384	–	–	–	1 (low)	4 (high)
user	categorical	4388	–	–	–	0 (no)	1 (yes)
eco awareness	metric	4387	3.79	4	0.80	1 (very negative)	5 (very positive)
technology affinity	metric	4388	3.36	3	0.89	1 (very negative)	5 (very positive)
trust in policymakers	metric	4382	2.85	3	0.91	1 (very negative)	5 (very positive)
mandatory connection	categorical	4388	–	–	–	0 (no mandatory connection)	1 (mandatory connection)
price regulation	categorical	4388	–	–	–	0 (regulated)	1 (liberalised)
ownership structure	categorical	4388	–	–	–	0 (mainly public)	1 (mixed)

(Note: For gender, there is also the category 'other' with 14 respondents.)

Table 11
Econometric results from the ordinal logistic regression models (own calculations, data from Breitschopf et al., 2023).

	Model 1			Model 2			Model 3		
	coefficient (error)	OR	p-value	coefficient (error)	OR	p-value	coefficient (error)	OR	p-value
Intercepts									
1 | 2	−3.85 (0.13)	0.02	0.000	−4.73 (0.16)	0.01	0.000	−4.38 (0.17)	0.01	0.000
2 | 3	−2.27 (0.10)	0.10	0.000	−3.12 (0.14)	0.04	0.000	−2.79 (0.15)	0.06	0.000
3 | 4	0.01 (0.09)	1.01	0.910	−0.76 (0.13)	0.47	0.000	−0.48 (0.14)	0.62	0.000
4 | 5	1.81 (0.10)	6.11	0.000	1.12 (0.13)	3.06	0.000	1.33 (0.14)	3.77	0.000
Socio-demographic factors, users and attitudes									
gender male	−0.01 (0.06)	0.99	0.802	−0.03 (0.06)	0.97	0.651	0.00 (0.06)	1.00	0.993
Age	0.24 (0.03)	1.27	0.000	0.22 (0.03)	1.25	0.000	0.23 (0.03)	1.26	0.000
education level 2	−0.12 (0.10)	0.88	0.226	0.02 (0.10)	1.02	0.860	−0.04 (0.10)	0.96	0.710
education level 3	0.00 (0.10)	1.00	0.995	0.15 (0.11)	1.16	0.174	0.06 (0.11)	1.06	0.554
education level 4	−0.07 (0.10)	0.93	0.466	0.17 (0.10)	1.19	0.095	−0.02 (0.10)	0.98	0.843
User	0.67 (0.06)	1.96	0.000	0.46 (0.07)	1.58	0.000	0.56 (0.07)	1.75	0.000
eco awareness	0.27 (0.03)	1.31	0.000	0.30 (0.03)	1.35	0.000	0.28 (0.03)	1.33	0.000
technology affinity	0.20 (0.03)	1.22	0.000	0.18 (0.03)	1.20	0.000	0.18 (0.03)	1.20	0.000
trust in policymakers	0.50 (0.03)	1.65	0.000	0.46 (0.03)	1.59	0.000	0.49 (0.03)	1.63	0.000
Country effects (Denmark is reference)									
France				−1.19 (0.13)	0.30	0.000			
Germany				−0.67 (0.12)	0.51	0.000			
Italy				−0.73 (0.13)	0.48	0.000			
Lithuania				−1.63 (0.13)	0.20	0.000			
Netherlands				−1.58 (0.13)	0.21	0.000			
Poland				−0.66 (0.12)	0.52	0.000			
Slovakia				−0.97 (0.13)	0.38	0.000			
Sweden				−0.52 (0.12)	0.59	0.000			
Regulation									
mandatory connection							−0.48 (0.07)	0.62	0.000
price regulation							−0.12 (0.07)	0.89	0.114
ownership structure							−0.37 (0.08)	0.69	0.000
N	4377			4377			4377		
AIC	11007			10762			10958		

(Note: Coefficients are scaled in terms of logs; OR = odd ratios, i.e. exponentiated coefficients)

Table 12
Econometric results from the ordinal logistic regression model 1 per country (own calculations, data from Breitschopf et al., 2023).

	Denmark		France		Germany		Italy		Lithuania		Netherlands		Poland		Slovakia		Sweden	
	coefficient (error)	p-value	coefficient (error)	p-value	coefficient (error)	p-value	coefficient (error)	p-value	coefficient (error)	p-value	coefficient (error)	p-value	coefficient (error)	p-value	coefficient (error)	p-value	coefficient (error)	p-value
Intercepts																		
1 | 2	−3.95 (0.42)	0.000	−4.06 (0.41)	0.000	−4.38 (0.52)	0.000	−5.31 (0.62)	0.000	−2.44 (0.62)	0.000	−4.08 (0.41)	0.000	−4.44 (0.61)	0.000	−3.44 (0.62)	0.000	−4.59 (0.50)	0.000
2 | 3	−2.94 (0.32)	0.000	−2.01 (0.30)	0.000	−2.58 (0.32)	0.000	−3.08 (0.29)	0.000	−1.25 (0.61)	0.039	−2.02 (0.34)	0.000	−3.05 (0.52)	0.000	−1.81 (0.56)	0.001	−2.86 (0.30)	0.000
3 | 4	−0.65 (0.26)	0.012	0.12 (0.29)	0.683	0.13 (0.27)	0.641	−0.22 (0.23)	0.339	0.75 (0.60)	0.215	0.26 (0.33)	0.422	−0.45 (0.50)	0.366	0.88 (0.55)	0.108	−0.04 (0.24)	0.852
4 | 5	1.25 (0.26)	0.000	2.23 (0.31)	0.000	2.10 (0.29)	0.000	1.87 (0.24)	0.000	2.38 (0.62)	0.000	2.37 (0.36)	0.000	1.45 (0.50)	0.004	2.49 (0.56)	0.000	1.98 (0.26)	0.000
Socio-demographic factors, users and attitudes																		
gender male	−0.06 (0.18)	0.729	−0.19 (0.18)	0.283	0.00 (0.18)	0.995	−0.16 (0.17)	0.375	0.28 (0.19)	0.141	−0.01 (0.18)	0.966	−0.08 (0.18)	0.638	−0.09 (0.18)	0.619	0.06 (0.18)	0.736
age	0.64 (0.10)	0.000	0.09 (0.09)	0.360	0.09 (0.09)	0.292	0.39 (0.09)	0.000	0.06 (0.10)	0.559	−0.3 (0.10)	0.003	0.31 (0.09)	0.000	0.03 (0.09)	0.788	0.52 (0.09)	0.000
education level 2	−0.07 (0.31)	0.825	0.25 (0.31)	0.418	0.29 (0.32)	0.365	0.27 (0.24)	0.254	−0.2 (0.64)	0.755	−0.62 (0.36)	0.085	−0.12 (0.49)	0.799	0.52 (0.55)	0.347	−0.26 (0.28)	0.339
education level 3	0.19 (0.26)	0.469	−0.18 (0.33)	0.574	0.37 (0.29)	0.204	0.46 (0.31)	0.135	0.07 (0.60)	0.902	−0.52 (0.35)	0.137	−0.28 (0.5)	0.581	0.05 (0.62)	0.931	0.10 (0.30)	0.725
education level 4	0.24 (0.27)	0.360	0.03 (0.30)	0.912	0.28 (0.30)	0.348	0.38 (0.28)	0.172	−0.15 (0.59)	0.806	−0.59 (0.35)	0.090	−0.12 (0.5)	0.806	0.75 (0.56)	0.187	0.48 (0.26)	0.062
User	0.79 (0.18)	0.000	0.38 (0.30)	0.202	0.90 (0.24)	0.000	−0.03 (0.33)	0.939	0.39 (0.18)	0.029	0.01 (0.35)	0.987	0.39 (0.18)	0.031	0.08 (0.22)	0.726	0.62 (0.17)	0.000
eco awareness	0.39 (0.09)	0.000	0.35 (0.09)	0.000	0.43 (0.10)	0.000	0.44 (0.10)	0.000	0.15 (0.10)	0.125	0.39 (0.09)	0.000	0.32 (0.10)	0.001	0.28 (0.10)	0.005	0.02 (0.09)	0.838
technology affinity	0.15 (0.10)	0.141	0.46 (0.10)	0.000	0.02 (0.10)	0.835	0.37 (0.09)	0.000	−0.11 (0.09)	0.229	0.04 (0.10)	0.658	0.20 (0.10)	0.038	0.11 (0.10)	0.297	0.39 (0.10)	0.000
trust in policymakers	0.46 (0.10)	0.000	0.46 (0.09)	0.000	0.33 (0.10)	0.001	0.36 (0.09)	0.000	0.34 (0.10)	0.000	0.64 (0.01)	0.000	0.50 (0.09)	0.000	0.44 (0.09)	0.000	0.58 (0.01)	0.000
N	500		490		491		512		449		475		503		464		493	
AIC	1135		1234		1176		1175		1264		1192		1220		1158		1140	

(Note: It should be noted that in France, Italy and the Netherlands the sample obtained includes only a small number of district heating users, which makes it difficult to obtain significant results on country-specific issues for users in these three countries.)

methodology in section 3). Since the regulation variables contain different country groups, the countries are covered in model 3 by the regulation variables and do not appear as separate predicting variables. Overall, the results of the econometric models are in line with the results of the group comparison and test statistics in sections 4.1, 4.2 and 4.3. Thus, the econometric results verify the tendencies identified based on the test statistics.

The coefficients of gender are not statistically significant in all models, thus, there are no differences between male and female respondents in the perception of district heating. In contrast, the coefficients of age are significantly positive in all models, indicating that higher age leads to a more positive perception of district heating. The coefficients of education, however, are not significant in any model. Further, the results show that being a user of district heating has an impact, as the coefficients of user are significantly positive in all models. In comparison, the coefficient of users reaches the highest effect in model 1 and 3, indicating that being a user has a high impact on the perception of district heating (compare odd ratios in Table 11).

Regarding attitudes, the results show that the predictors eco awareness, technology affinity and trust in policymakers have significantly positive coefficients in all models, indicating that they are important predictors of the perception of district heating. Hence, higher eco awareness, higher affinity to (new) technologies and greater trust in policymakers leads to a better perception of district heating. Thereby, trust in policymakers shows the highest impact of these tree predictors.

Furthermore, model 2 shows that the perception is significantly impacted by the country. Thereby the results on the perception of district heating in the regression are in line with the test statistics from section 4.1. The biggest difference can be observed between Denmark and Lithuania, followed by the Netherlands and France (compare coefficients in Table 11). Moreover, model 2 achieves the best fitting of all three models, i.e. the smaller the AIC, the better the model fit (compare AIC in Table 11).

Analysing the regulation variables, i.e. mandatory consumer connection, price regulation, and ownership structure, the results of model 3 show that mandatory connection and ownership structure are significant with a negative direction, i.e. the coefficients show that no mandatory connection and mainly public ownership support a positive perception of district heating. Price regulation (only) shows a p-value of 0.114 and is therefore not significant.

In addition to the three models using the full sample, model 1 was calculated separately for each country (compare Table 12). There are some similarities between the countries, but also clear differences. In all countries, the coefficients of gender and education are not statistically significant. Age shows significant coefficients (only) in Denmark, Italy, the Netherlands, Poland and Sweden. Being a user shows significant coefficients in the majority of countries, i.e. Denmark, Germany, Lithuania, Poland and Sweden. Furthermore, the coefficients of eco awareness are significantly positive in all countries except Lithuania and Sweden. Technology affinity is (only) relevant in some countries, while greater trust in policymakers leads to a better perception of district heating in all countries. Looking at the countries with the most positive and most negative perceptions, there are some interesting differences. The results show that in Denmark eco awareness has a significant impact, while in Sweden technology affinity is of greater relevance. In the Netherlands, trust in policymakers shows the highest significant coefficients and in Lithuania, being a user and (also) high trust in policymakers are relevant. Overall, the results are in line with the previous findings from section 4.2.

5. Discussion

The results show that the majority of the 4388 European respondents have an overall positive perception of district heating. In Denmark and Sweden, the perception of the respondents is particularly positive. In the Netherlands and Lithuania, the perception of district heating is less positive. In addition, the results of the analyses show that attitudes, values and trust, i.e. eco awareness, technology affinity and trust in policymakers, have a stronger influence on the perception than socio-demographic factors, i.e. gender, age and education. Gender and education have no impact in any model on the perception of district heating. In contrast, the results indicate that higher age leads to a more positive perception of district heating. Furthermore, being a user of district heating shows a positive impact on the perception of district heating in all models. This result is not overly surprising given the fact that users have more experience with district heating as a heating system than non-users. Following research based on the diffusion on innovation theory by (2003) Rogers, the trialability of a (new) technology as well as the compatibility of the technology (here: district heating) increases the likelihood of adopting innovations and their rating, as has already been shown for other technologies, e.g. electric vehicles (Preuß et al., 2022) or car sharing (Burger et al., 2020).

Comparing countries, there are some clear differences: For example, in Denmark, the eco awareness of respondents seems to be very relevant, while in Sweden technology affinity seems to have a stronger impact on the perception of district heating. In contrast, in Lithuania and the Netherlands, trust in policymakers reveals high effects. The positive impact of eco awareness on the perception of district heating could be connected to the share of renewables. Denmark has very high shares of renewables in their district heating mix (around 60%; Bacquet et al., 2022). Households might recognise the high share of renewables in district heating, especially households with a high eco awareness, contributing to the (overall) positive perception. A connection between perception and climate-friendly (district) heating is also established by other studies (e.g. Krikser et al., 2020; Zaunbrecher et al., 2016). Thus, our results indicate that increasing renewables in district heating is not only a means to reach climate targets but also leads to a better perception.

Regarding regulations and institutional settings, the results show that respondents from countries without mandatory connection have a more positive perception and a higher satisfaction with district heating as well as a more positive rating of their district heating price. Households with mandatory grid connections may feel locked in, as they neither have a free choice for their heating system nor for the heat supplier, which might explain why the perception is less positive (compare also Hodges et al., 2018). However, Denmark and Germany show a positive perception despite having mandatory consumer connections. Furthermore, respondents from the countries with more liberalised price regulation show a more positive perception, a higher level of satisfaction with district heating and a more positive rating of their price than respondents from countries with more regulated prices. Again, exceptions exist for households from some countries with regulated prices (e.g. Denmark), which also show a positive perception, satisfaction and rating of its price. These exceptions may show that it is not the formal regulation that is relevant, but how successful the regulation is, in terms of actually leading to reasonable and transparent prices and limiting the market power of district heating suppliers. Lastly, regarding ownership, the results show that public ownership of district heating seems to positively impact consumer perception, satisfaction and price rating. Public ownership of district heating leads to a greater ability for households to participate and influence decisions regarding their local district heating system, which most likely explains the better perception.

Overall, our results show that respondents from countries with no mandatory connection, liberalised price regulation and mainly public ownership seem to have a more positive perception of district heating, higher satisfaction when using district heating and a more positive rating of their heating price. However, there are exceptions and further research should focus on the impact and interplay of regulations that may influence consumers' perception of district heating, building on the results shown in this paper.

6. Conclusions and policy implications

Europe aims to increase the deployment of district heating, i.e. at least doubling its market share by 2050 (compare section 1). As this transition directly affects end-users, their perception is key to further promoting the transition. This paper focuses on households' perception of district heating in nine European countries: Denmark, France, Germany, Italy, Lithuania, Netherlands, Poland, Slovakia and Sweden. The analysis uses data from an online survey with 4388 participants.

Overall, the research question of this paper can be answered to the extent that the selected socio-demographic factors and attitudes of the respondents, as well as the usage of district heating, are linked to their perception of district heating. Older age and being a user as well as having a higher eco awareness, a higher affinity to (new) technologies and a greater trust in policymakers leads to a more positive perception (than low levels of these variables).

In addition, the country comparisons show that respondents from Denmark and Sweden have a more positive perception while respondents from Lithuania and the Netherlands have a less positive perception of district heating. These country differences are most likely explained by a variety of aspects. Some of the reasons are beyond the scope of this paper such as historical developments and experiences, technical specifications and the efficiency of existing district heating networks. However, our results show that attitudes seem to have a stronger influence on the perception than socio-demographic factors in (almost) all countries.

In particular, eco awareness shows a significant positive effect in the majority of countries. Thus, we conclude that policy measures that address the awareness of environmental benefits can help to increase positive perceptions and in turn the deployment of district heating. Furthermore, greater trust in policymakers is linked to a better perception of district heating in all models and countries. Therefore, we deduce that policymakers need to build trust in order to achieve a positive perception of district heating.

Finally, the results indicate that the mix of appropriate regulations and, in particular, their combined impact, could play an important role in perception. Respondents from countries with no mandatory connection, with liberalised price regulation and mainly public ownership, reveal a more positive perception of district heating than those from countries with mandatory connection, regulated prices and a more mixed ownership structure. However, there are exceptions and respondents from some countries with mandatory connection, regulated prices and more private ownership can also show a positive perception.

Therefore, we come to the final conclusion that not the formal regulation but the interplay of regulations and institutional settings ensuring clarity of and compliance with pricing rules, restriction of market power, transparency of markets as well as some form of consumer empowerment or participation, are key elements for a positive perception of district heating. Overall, the paper provides a first overview of possible factors influencing the perception of district heating and indicates that regulations and their interplay are relevant drivers, which should be investigated in further research.

CRediT authorship contribution statement

Anna Billerbeck: Conceptualization, Methodology, Software, Formal analysis, Investigation, Resources, Visualization, Writing – original draft. **Barbara Breitschopf:** Methodology, Validation, Writing – review & editing. **Sabine Preuß:** Methodology, Validation, Writing – review & editing. **Jenny Winkler:** Writing – review & editing. **Mario Ragwitz:** Supervision. **Dogan Keles:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix

Table A1
Overview of questions from the online survey used in this paper (Breitschopf et al., 2023).

Variable	Question and answer options
perception	Question: How do you rate district heating as a heating system overall? Answer: (1) very negative, (2) negative, (3) neutral, (4) positive, (5) very positive
satisfaction	Question: How satisfied are you with your current heating system? Answer: (1) very dissatisfied, (2) rather dissatisfied, (3) neutral, (4) rather satisfied, (5) very satisfied
price rating	Question: How would you describe your price for heating with district heating (in comparison with your income)? Answer: (1) very high, (2) high, (3) neutral, (4) low, (5) very low
gender	Question: Please indicate your gender. Answer: Female, Male, None of the above, I do not want to answer
age	Question: How old are you? Answer: [year]
education	Question: Which is the highest level of education that you have completed? Answer: Primary level, Secondary level (collage, high school, middle school), Vocational/technical training or education, Academic degree (Bachelor and Master degree or PhD)
user	Question: What heating system do you currently use in your home, i.e. how do you heat your rooms? Answer: Several options, including district heating
eco awareness	Question: To what extent do you agree or disagree with the following statements? (1) Acting environmentally-friendly is an important part of who I am (2) I am the type of person who acts environmentally-friendly (3) I see myself as an environmentally-friendly person Answer: (1) I totally agree, (2) I agree, (3) neutral, (4) I disagree, (5) I totally disagree
technology affinity	Question: To what extent do you agree or disagree with the following statements? (1) I am very interested in the latest technology developments (2) It does not take me long to learn to like new technology developments (3) I am always keen to use the latest technological device Answer: (1) I totally agree, (2) I agree, (3) neutral, (4) I disagree, (5) I totally disagree
trust in policymakers	Question: How much do you trust the following actors in your country to make good decisions regarding district heating? (1) Regional government, (2) National government, (3) European commission Answer: (1) not at all to (5) completely

Table A2
Overview of sample and quotas (country shares based on Eurostat and Bacquet et al., 2022).

Country	Share DH users		Share female		Average age		Share tertiary education		Average house-hold size	
	sample	country	sample	country	sample	country	sample	country	sample	country
Denmark	57%	65%	50%	50%	53	42	68%	47%	1.93	2.0
France	10%	8%	50%	52%	54	42	59%	49%	2.23	2.3
Germany	15%	15%	50%	51%	56	46	69%	35%	1.99	2.0
Italy	7%	6%	51%	51%	48	47	35%	29%	2.29	2.3
Lithuania	41%	57%	66%	53%	45	44	86%	56%	2.55	2.2
Netherlands	8%	6%	52%	50%	52	43	67%	52%	2.12	2.1
Poland	39%	43%	57%	52%	41	41	53%	42%	2.73	2.8
Slovakia	22%	34%	54%	51%	43	41	34%	39%	2.93	2.9
Sweden	45%	50%	52%	50%	54	41	56%	49%	2.01	2.0

Table A3
Satisfaction with the use of district heating in the different countries (own calculations; data from Breitschopf et al., 2023).

Country	n (users)	mean	median	Rating				
				Very dissatisfied	rather dissatisfied	neutral	rather satisfied	Very satisfied
Denmark	284	3.70	4	7%	6%	24%	33%	29%
France	48	3.60	4	2%	19%	15%	46%	19%

(continued on next page)

Table A3 (*continued*)

Country	n (users)	mean	median	Rating				
				Very dissatisfied	rather dissatisfied	neutral	rather satisfied	Very satisfied
Germany	73	3.78	4	5%	8%	21%	34%	32%
Italy	38	3.53	4	8%	11%	18%	47%	16%
Lithuania	183	3.28	4	7%	19%	27%	34%	14%
Netherlands	36	3.31	4	8%	19%	19%	39%	14%
Poland	195	3.79	4	6%	8%	5%	62%	19%
Slovakia	102	3.35	4	10%	13%	25%	36%	16%
Sweden	222	3.85	4	1%	9%	20%	45%	26%

Table A4
Mean and median of perception in the context of socio-demographic factors and attitudes per country(own calculations; data from Breitschopf et al., 2023).

Country	Variable		mean	median
Denmark	gender	male	4.04	4
		female	4.02	4
	age	older	4.26	4
		younger	3.77	4
	education	higher	4.14	4
		lower	3.81	4
	user	user	4.17	4
		non-user	3.85	4
	eco awareness	higher	4.22	4
		lower	3.71	4
	technology affinity	higher	4.08	4
		lower	3.98	4
	trust in policymakers	higher	4.24	4
		lower	3.88	4
France	gender	male	3.41	3
		female	3.41	3
	age	older	3.43	3
		younger	3.39	3
	education	higher	3.38	3
		lower	3.46	3
	user	user	3.46	3
		non-user	3.41	3
	eco awareness	higher	3.59	4
		lower	3.07	3
	technology affinity	higher	3.72	4
		lower	3.17	3
	trust in policymakers	higher	3.57	4
		lower	3.19	3
Germany	gender	male	3.65	4
		female	3.70	4
	age	older	3.75	4
		younger	3.58	4
	education	higher	3.70	4
		lower	3.62	4
	user	user	3.99	4
		non-user	3.63	4
	eco awareness	higher	3.82	4
		lower	3.43	3
	technology affinity	higher	3.75	4
		lower	3.60	4
	trust in policymakers	higher	3.78	4
		lower	3.53	4
Italy	gender	male	3.69	4
		female	3.73	4
	age	older	3.81	4
		younger	3.60	4
	education	higher	3.79	4
		lower	3.67	4
	user	user	3.66	4
		non-user	3.72	4
	eco awareness	higher	3.90	4
		lower	3.50	3
	technology affinity	higher	3.86	4
		lower	3.56	4
	trust in policymakers	higher	3.78	4
		lower	3.60	4
Lithuania	gender	male	3.31	3
		female	3.15	3

(*continued on next page*)

Table A4 (*continued*)

Country	Variable		mean	median
	age	older	3.23	3
		younger	3.18	3
	education	higher	3.21	3
		lower	3.17	3
	user	user	3.33	3
		non-user	3.12	3
	eco awareness	higher	3.27	3
		lower	3.12	3
	technology affinity	higher	3.19	3
		lower	3.22	3
	trust in policymakers	higher	3.37	3
		lower	2.98	3
Netherlands	gender	male	3.15	3
		female	3.15	3
	age	older	3.00	3
		younger	3.31	3
	education	higher	3.18	3
		lower	3.10	3
	user	user	3.11	3
		non-user	3.16	3
	eco awareness	higher	3.29	3
		lower	2.92	3
	technology affinity	higher	3.20	3
		lower	3.09	3
	trust in policymakers	higher	3.36	3
		lower	2.88	3
Poland	gender	male	3.67	4
		female	3.75	4
	age	older	3.84	4
		younger	3.61	4
	education	higher	3.71	4
		lower	3.72	4
	user	user	3.82	4
		non-user	3.65	4
	eco awareness	higher	3.83	4
		lower	3.51	3
	technology affinity	higher	3.82	4
		lower	3.58	3
	trust in policymakers	higher	3.81	4
		lower	3.58	4
Slovakia	gender	male	3.38	3
		female	3.47	3
	age	older	3.44	3
		younger	3.41	3
	education	higher	3.48	3
		lower	3.39	3
	user	user	3.44	4
		non-user	3.42	3
	eco awareness	higher	3.55	3
		lower	3.27	3
	technology affinity	higher	3.56	4
		lower	3.32	3
	trust in policymakers	higher	3.55	3
		lower	3.23	3
Sweden	gender	male	3.81	4
		female	3.71	4
	age	older	3.87	4
		younger	3.63	4
	education	higher	3.86	4
		lower	3.60	3
	user	user	3.91	4
		non-user	3.62	4
	eco awareness	higher	3.87	4
		lower	3.60	3
	technology affinity	higher	3.95	4
		lower	3.58	4
	trust in policymakers	higher	3.84	4
		lower	3.59	4

Table A5
Regulations and ownership structure in the nine countries.

Country	Regulation
Denmark	Denmark applied mandatory grid connection for consumers in the past. Municipalities were able to impose a mandatory connection when it proves to be the most cost-efficient heating solution (Bacquet et al., 2022). In 2019, this regulation was discarded, however, the existing obligations, which affect around half of all district heating consumers in Denmark, remained in place (Bacquet et al., 2022; Euroheat & Power, 2019). At the same time, the Danish district heating sector is designed as non-profit and legislation has a strong consumer orientation (Bacquet et al., 2022). Regarding price regulation, Denmark follows the 'true-cost principle', i.e. prices can only cover necessary costs but, with a few minor exemptions, not a profit (Bacquet et al., 2022). In addition, a price cap for waste heat was introduced, which came into force in January 2022 (Horten, 2022). Looking at the ownership structure, only around 5% are in private ownership, underlining the non-profit setting (Bacquet et al., 2022). Municipalities own around 60% of the district heating networks and around 35% are owned by consumer cooperatives (Bacquet et al., 2022).
France	France follows a more liberalised approach, compared to Denmark. They rely on mandatory connection under certain conditions. New and renovated buildings located within a classified area are obliged to be connected to district heating (Bacquet et al., 2022). Furthermore, the 'private operation under public ownership' model is quite common. If a private company operates a public network, prices are negotiated with the local authority and fixed in an agreement (Bacquet et al., 2022). Prices can be adjusted following specific indexing formulas and revised in certain circumstances set in the agreement (Bacquet et al., 2022). In line, around 44% of the district heating networks in France are operated in a public-private partnership, 37% are owned by public companies and around 18% are private (Bacquet et al., 2022).
Germany	In some municipalities in Germany, consumers must connect to their local district heating grid. Municipalities can determine by statute that within a certain area, each property must be connected (Bacquet et al., 2022). Regarding pricing, Germany relies on price adjustment clauses, i.e. prices orientated on energy carrier prices, mostly gas and coal (Bacquet et al., 2022). Hence, the 'substitution price' model (more or less) describes the price regulation. In light of the current gas crisis, price caps for gas, electricity and district heating were implemented in 2023, i.e. after the household survey (Bundesregierung, 2023). The district heating price cap applies to 80% of the heat consumed, while the remaining 20% of the consumption is not covered to create incentives to save energy. The cap is set by 9.5 ct/kWh (Bundesregierung, 2023). This more liberalised price regulation is combined with mainly public ownership of the existing district heating networks (Bacquet et al., 2022).
Italy	Italy does not rely on mandatory connections for consumers (Bacquet et al., 2022). At the same time, price regulation is more liberalised and more or less in line with the 'substitution price' model. Prices are calculated by operators based on a sustained cost criterion or based on the avoided cost, i.e. the cost that the user would have sustained using a different technology (Bacquet et al., 2022). Regarding ownership, most of the systems were originally publicly owned, however, newly developed systems are mostly privately owned and operated (Bacquet et al., 2022). The three large district heating networks supplying Turin, Brescia and Milan are operated under public-private partnership (Bacquet et al., 2022).
Lithuania	Lithuania follows overall a more regulated approach. All district heating operators must hold a license and their activities, including pricing, are controlled by the regulatory authority (Bacquet et al., 2022). For all licensed areas, grid connection is obligatory for consumers (Bacquet et al., 2022). Prices are set on a cost-plus basis and the regulatory authority approves the methodologies that companies apply for calculating prices (Bacquet et al., 2022). Further, district heating with at least one independent producer must hold a monthly auction, determining which producer will cover the forecasted heat demand (Blömer, R., Billerbeck, A., Winkler, J., 2022). The auctions impact the end consumer price as they are cleared based on the lowest prices. Regarding ownership, the networks are primarily in public ownership, i.e. municipalities own about 93% of all district heating companies (Bacquet et al., 2022).
Netherlands	In the Netherlands, there are very few cities with mandatory consumer connections (existing district heating systems) (Bacquet et al., 2022). Furthermore, under a new law, municipalities should develop so-called environmental plans with district heating zoning (i.e. district heating as default. However, there is always an opt-out for consumers (Klimaatweb, 2023). The Netherlands also has a more regulated pricing mechanism. The regulatory authority sets tariffs each year based on the price of natural gas (Bacquet et al., 2022). Hence, also the Netherlands relies on a 'substitution price'. Though the regulator sets the price and not the operator. Similar to Germany, in light of the gas crisis, a price cap was introduced in 2023 (Government of the Netherlands). This cap is set at 47.38 €/GJ and it is also limited to a certain consumption, i.e. 37 GJ of district heating used (Government of the Netherlands). The ownership structure is more mixed compared to the other countries. Around 57% of the networks are publicly owned or owned by consumer cooperatives, while 43% are privately owned (Bacquet et al., 2022). Thereby, the five largest networks are owned by private companies (Bacquet et al., 2022).
Poland	Poland does not rely on mandatory connections for consumers (Bacquet et al., 2022). Consumer access to the grid is regulated with detailed conditions, i.e. operators are obliged to arrange a fair grid connection agreement with consumers requesting connection to the grid, if technically and economically feasible, however, there are no obligations for consumers to connect to and use their local district heating network (Bacquet et al., 2022). At the same time, prices are more regulated. Polish district heating companies prepare prices and the regulatory authority accepts or rejects them (Bacquet et al., 2022). Thereby, the prices are set with a function of justified costs and plans of investments (Bacquet et al., 2022). Hence, the 'true-cost principle' can be used to describe the price regulation in place. The ownership structure in Poland is more mixed with nearly 50% of district heating companies being controlled by public operators and the majority of big networks in large cities being controlled by the private sector (Bacquet et al., 2022).
Slovakia	In Slovakia, larger consumers must connect to district heating (Bacquet et al., 2022). This mandatory connection is combined with a more regulated approach to pricing. Prices are regulated by the application of a price calculation rule, which is based on the calculation of economically eligible costs and a fair profit (Bacquet et al., 2022). Hence, the regulation is similar to the 'true-cost principle', with the exception that the inclusion of a profit is allowed. In addition, the regulatory office sets price caps for fuels (natural gas, biomass, coal) that the suppliers are eligible to include in the heating price. Regarding ownership, the state owns and operates large district heating networks in six cities (Bratislava, Košice, Žilina, Trnava, Zvolen and Martin) (Bacquet et al., 2022).Private players own all the district heating networks except for these networks (Bacquet et al., 2022).
Sweden	The Swedish district heating market is overall more liberalised and there is no mandatory grid access for consumers (Bacquet et al., 2022). In the past, district heating was deregulated and they followed the 'prices set by the market' option (Odgaard and Djørup, 2020). However, due to limited competition on the heat production side, prices increased strongly and thus the pricing regime was reformed in 2012 (Odgaard and Djørup, 2020). Today, no formal price regulation but a negotiation mechanism is in place that obliges district heating companies to submit their operating reports to the regulators and provide pricing information. In addition, the companies must ensure that information on the prices for district heating is clear and correct and readily available to customers and the general public (Bacquet et al., 2022). At the same time, district heating in Sweden is mainly in public ownership, i.e. at least 65% of the networks are owned and operated by public entities (Bacquet et al., 2022).

Table A6
Correlations between predictors (own calculations; data from Breitschopf et al., 2023).

variables	age	user	education	eco awareness	technology affinity	trust in policymakers	Price regulation	mandatory connection	ownership structure
age	1								
user	−0.02 p = 0.308	1							
education	−0.06 p = 0.000	0.06 p = 0.000	1						
eco awareness	0.05 p = 0.000	−0.05 p = 0.000	0.03 p = 0.026	1					

(continued on next page)

Table A6 (*continued*)

variables	age	user	education	eco awareness	technology affinity	trust in policymakers	Price regulation	mandatory connection	ownership structure
technology affinity	−0.20 p = 0.000	0.00 p = 0.856	0.06 p = 0.000	0.30 p = 0.000	1				
trust in policy makers	−0.11 p = 0.000	0.02 p = 0.234	0.06 p = 0.000	0.18 p = 0.000	0.22 p = 0.000	1			
price regulation	0.14 p = 0.000	−0.16 p = 0.000	−0.10 p = 0.000	0.08 p = 0.000	0.01 p = 0.783	0.05 p = 0.001	1		
mandatory connection	0.06 p = 0.103	−0.05 p = 0.003	0.12 p = 0.000	−0.05 p = 0.808	−0.15 p = 0.000	−0.08 p = 0.000	−0.31 p = 0.000	1	
ownership structure	−0.15 p = 0.000	−0.16 p = 0.000	−0.13 p = 0.000	0.08 p = 0.000	0.11 p = 0.000	−0.02 p = 0.256	−0.35 p = 0.000	−0.33 p = 0.000	1

(Note: Gender is not included in this table, but still used as a predictor in the econometric model.)

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